

TrustSyn — Biologically informed drug-combination prediction

Explicit uncertainty, novelty detection, and model accountability for prioritizing follow up experiments.



Presented by OncoNova

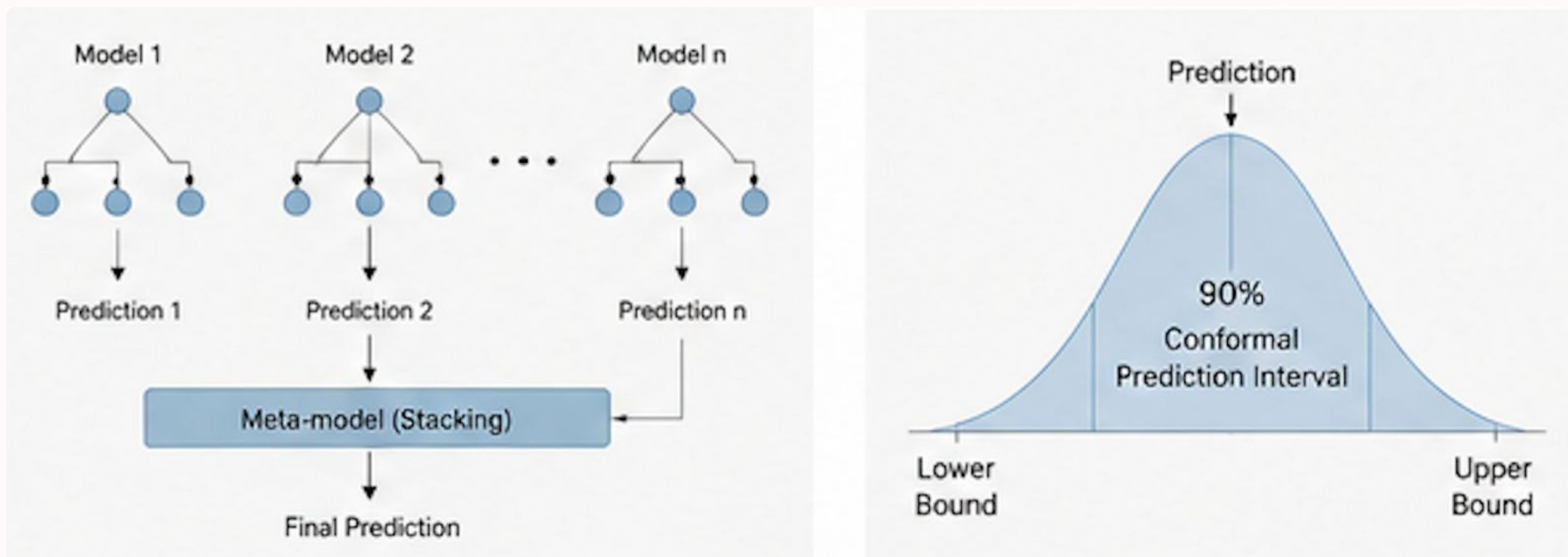
Team: OncoNova · System: TrustSyn · Task: Prioritize drug combinations from NCI -ALMANAC with cell -line context from CellMiner

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The practical problem

TrustSyn reframes model outputs into actionable prioritization by combining predictive scores with calibrated uncertainty, novelty detection, and explainability.



From left to right, model scores are filtered by trust signals so researchers can decide which combinations are worth testing next.

- Data sources: NCI -ALMANAC combo_score; CellMiner molecular features.
- Risk: IID metrics can mask failures on unseen drugs, combinations, or cell lines.
- Solution: evaluate four canonical splits + trust signals + explanations.

Four evaluation regimes (interpolation → extrapolation)

From standard IID evaluation to the most demanding generalization setting, each regime removes more familiar structure and increases difficulty.

TrustSyn reports performance per regime and attaches trust annotations to individual rows to guide experimental prioritization.

Regime definitions

- **RANDOM** — standard IID split (interpolation baseline).
- **COLD_COMBINATION**— hold out specific drug pairs (unseen combos).
- **COLD_CELL_LINE**— hold out cell lines (unseen biological contexts).
- **COLD_DRUG**— hold out drugs entirely (strongest extrapolation).



Modeling stack: complementary, split-aware

Diverse inputs

NCI -ALMANAC assay scores +
CellMiner molecular PCs, pathway
and target annotations.

Core models

D MPNN V2 C (graph neural
network), CatBoost / XGBoost,
LightGBM, and Ridge.

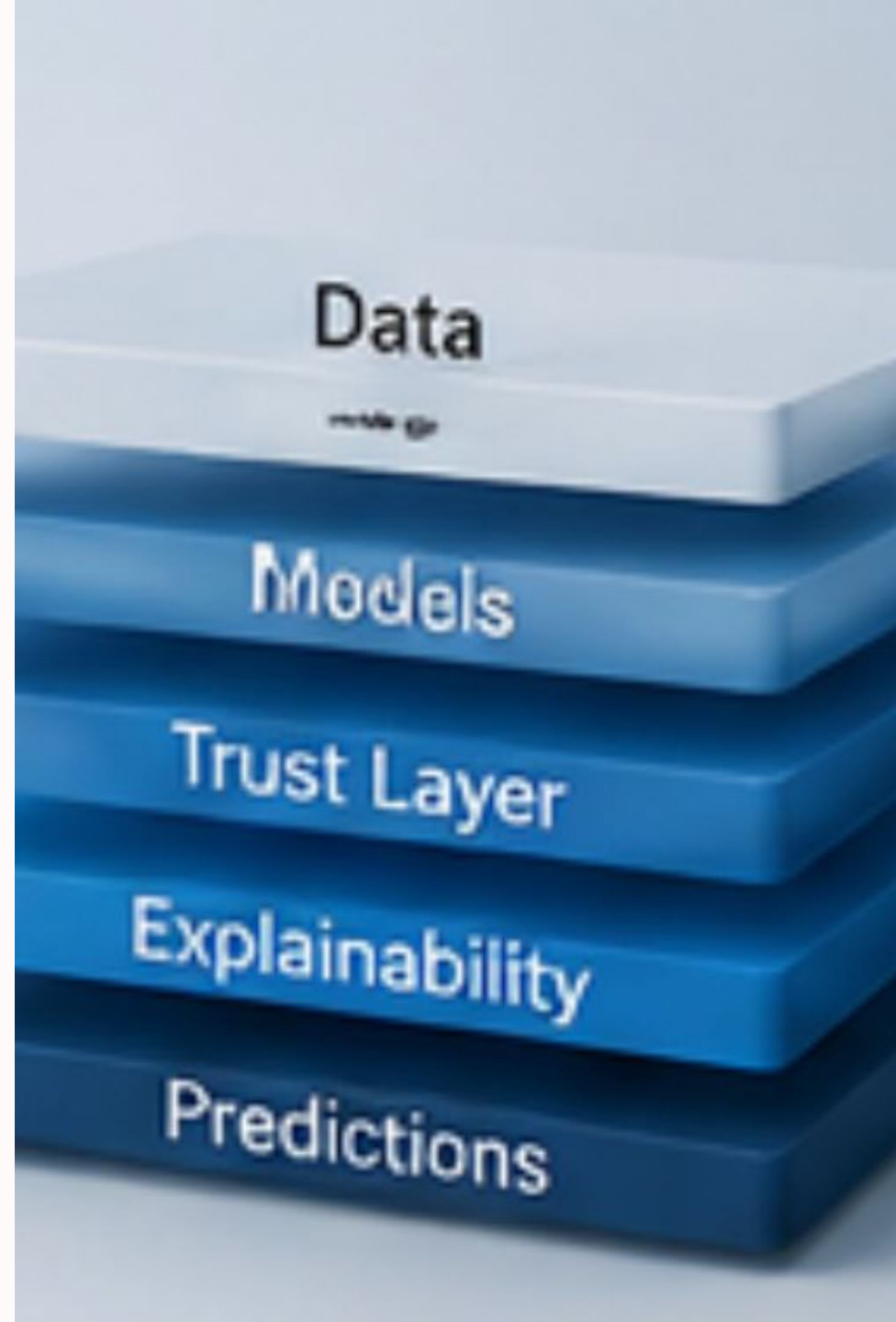
Split-specific stacking

Meta models chosen per split:
Ridge (RANDOM), XGBoost
(COLD_COMBINATION &
COLD_CELL_LINE), dedicated
stacking artifact for COLD_DRUG.

Trust layer + explanation

Agreement ensembles, conformal
intervals, novelty scores, and
SHAP based attributions produce
accountable outputs.

D MPNN V2 C config: hidden=128 · depth=2 · dropout=0.1 · LR=0.001 · batch=256 · seed=42.





Frozen predictive results (artifact evaluation)

Final metrics reported from supplied frozen artifacts (RMSE and R^2 per packaged split):

RANDOM · Ridge

RMSE 5.888 · R^2 0.294

COLD_COMBINATION · XGBoost

RMSE 6.568 · R^2 0.129

COLD_CELL_LINE · XGBoost

RMSE 6.789 · R^2 0.323

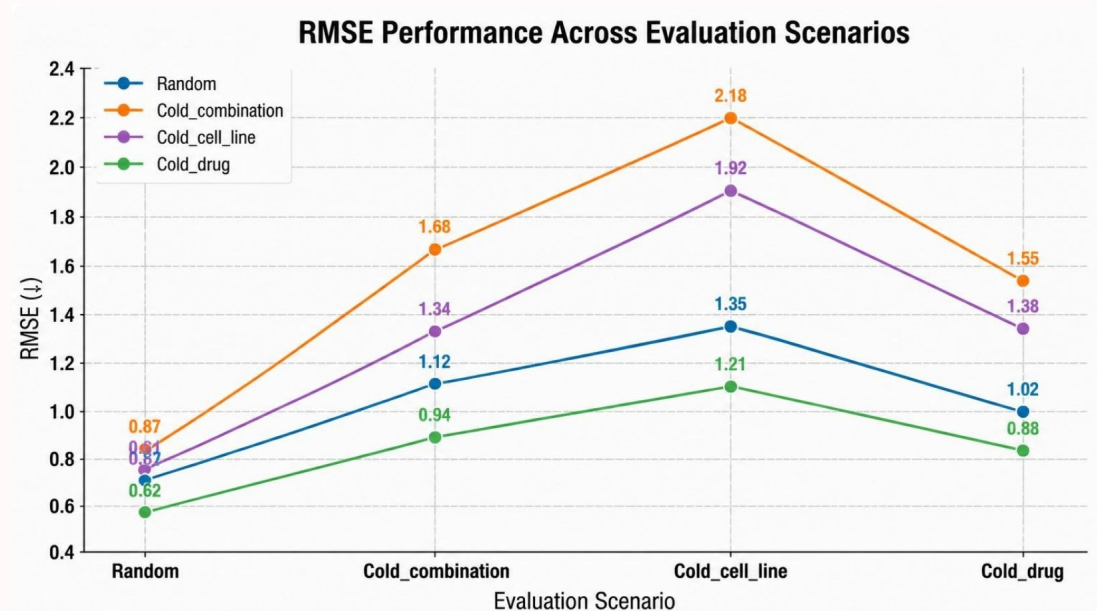
COLD_DRUG · dedicated stack

RMSE 4.858 · R^2 0.351 (separate
packaged artifact)

Interpretation: different splits favor different algorithms — motivating split aware stacking — rather than a single global winner.

D-MPNN V2-C: predictive strengths and failure modes

Evidence: D MPNN V2 C shows strong regression and ranking performance for interpolation and some extrapolation settings, but a pronounced ranking failure on COLD_DRUG (Precision@100 $\approx$ 0.05; nDCG@100 negative), indicating poor generalization to unseen chemotypes without dedicated handling.=

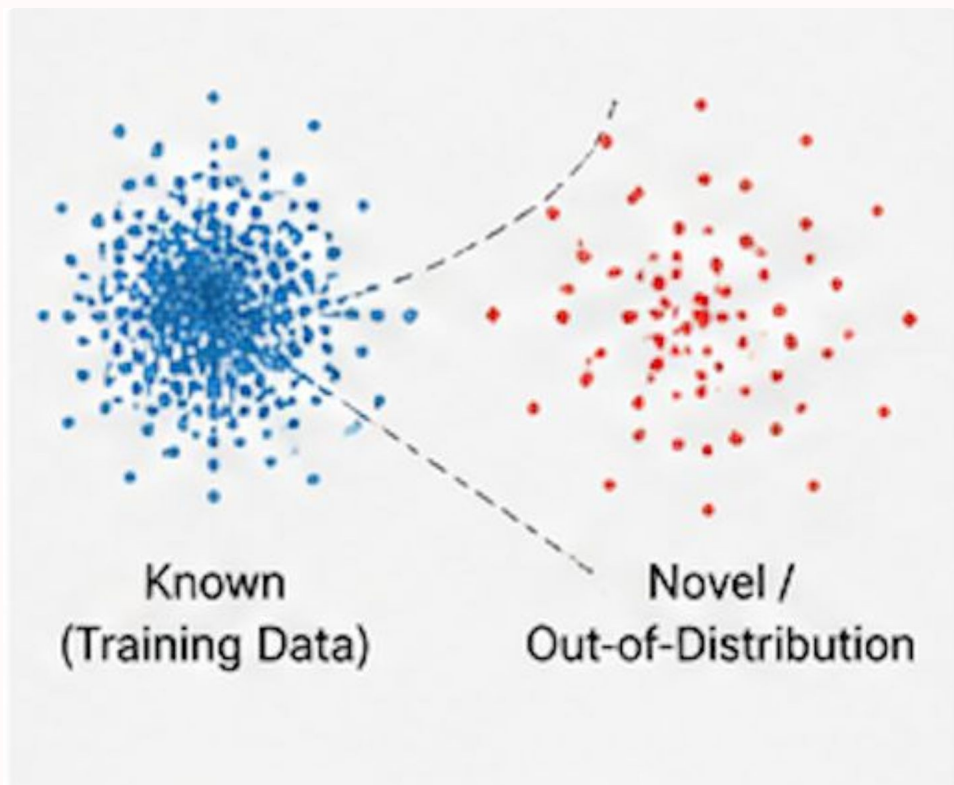




Trust layer: quantify when to be cautious

Validation summary across 88,753 packaged rows (RANDOM, COLD_COMBINATION, COLD_CELL_LINE): nominal 90% conformal coverage observed $\approx$ 90.7%, 90.8%, 88.5% respectively. High trust rows consistently exhibit lower observed MAE than low trust rows in packaged splits. Note: trust layer was not validated for COLD_DRUG (outside packaged scope).

Explainability — from score to evidence



Global CatBoost SHAP summary identifies the features driving model predictions:

- **union_target_count**— size of combined target set for the drug pair (biological plausibility).
- **Tanimoto_similarity**— chemical similarity between partner drugs (chemo-structural signal).
- **target_count_A / target_count_B**— per-drug polypharmacology.
- **STRING_available / KEGG_overlap** network and pathway context availability.
- **CellMiner_PC2, PC24**— cell-line molecular state captured by principal components.

Caveat: SHAP provides attribution for the model, not mechanistic causation. Use attributions to prioritize mechanistically plausible combinations for validation.

Key learnings — clear claims and disciplined limits

- Failure modes are real and informative: COLD_DRUG exposes D MPNN ranking failure (Precision@100 0.05; nDCG@100 -0.083).
- Novelty does not uniformly increase error — the novelty/error relationship is split dependent.
- Trust layer is validated on packaged splits only; COLD_DRUG requires separate trust validation.
- Frozen stacking artifact contains three packaged splits; COLD_DRUG is a separate packaged artifact.
- Positioning: TrustSyn is a prioritization and decision support system for guiding experiments — not a substitute for laboratory validation.



The final proposition

Predict across interpolation & extrapolation

Report split specific performance so stakeholders understand generalization boundaries.

Combine complementary models

Use split aware stacking rather than relying on a single architecture.

Attach uncertainty & trust

Conformal intervals + agreement + novelty scores support safer prioritization.

Expose feature evidence

SHAP attributions surface biological context to inform experimental design.

Be auditable & transparent

Publish frozen artifacts, split definitions, and trust diagnostics before experiments.

OncoNova • TrustSyn – an auditable prioritization workflow to guide preclinical combination experiments with explicit uncertainty and accountable explanations.

MANDATORY INTEGRITY SLIDE

OncoNova team · TrustSyn model

1. Generalization	COLD_DRUG is the strongest stress test for D-MPNN ranking. Its dedicated stacked artifact reports RMSE 4.858 and R^2 0.351.
2. Trust-layer scope	The supplied TrustSyn trust-layer CSV contains RANDOM, COLD_COMBINATION and COLD_CELL only. COLD_DRUG is not claimed as trust-layer validated.
3. Stacking scope	The frozen ensemble CSV contains RANDOM, COLD_COMBINATION and COLD_CELL. The COLD_DRUG stacked result is kept distinct and sourced from its dedicated JSON artifact.
4. Conformal prediction	Nominal 90% intervals show empirical coverage of approximately 90.7% RANDOM, 90.8% COLD_COMBINATION and 88.5% COLD_CELL.
5. Novelty	Novelty was investigated as an extrapolation-risk signal. Its association with error is not consistently positive across supplied splits; it is not presented as universally validated error prediction.
6. Biological interpretation	SHAP identifies influential chemical, target/network and CellMiner features. Attribution is not proof of mechanism; biological hypotheses require experimental validation.
7. Intended use	TrustSyn is a prioritization aid for experimental investigation. It does not replace laboratory validation or establish clinical efficacy.

Submission principle: measured results, interpretation and limitations are explicitly separated. No material limitation is hidden to improve the narrative.

AI assistance disclosure

ChatGPT (OpenAI) was used throughout the project as an AI-assisted development and documentation tool, including for code drafting and debugging support, data-analysis workflow assistance, technical writing, documentation, presentation development, and critical review of implementation logic.

The team remained responsible for scientific reasoning, methodological decisions, model development and evaluation, interpretation of results, verification of reported metrics, and final submission decisions. AI-generated suggestions were reviewed and validated against project artifacts before inclusion.